TFPT — Red Team / Stress Test layer

Five adversarial targets (A–E): try to *break* the reductions, not confirm them

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What this note is

This is the deliberately *adversarial* layer requested in the v78 review. Its purpose is **not** to confirm TFPT but to attack the five load-bearing reductions at their weakest logical transitions. Each target is run through one fixed protocol — minimal statement, assumptions, logical chain, validity conditions, counterexample search, limiting/degenerate cases, alternative structures, provisional verdict — and produces a deliberately narrow output: where the statement works, where it could fail, and the residual risk. The layer is machine-backed by [`verification/redteam/rt_A_e8net.py`] . . . [`verification/redteam/rt_E_vgeo.py`] and [`verification/redteam/run_redteam.py`]; a red-team check asserts an *adversarial* fact (a counterexample really exists, a hidden assumption is really needed, a firewall really holds), and the honest outcome lives in the **status** of each target, never in a green pass.

Target	Minimal claim	Status	Residual risk
A	seam–Calderón measure = $(E8)_1$ net	**reduced, not closed**	holomorphy proof + constructive map + bulk uniqueness
B	$g_{\mathrm{car}} = 5$ forced by the Pascal rule	**survives (narrowed)**	boundary derivation of half-spinor exhaustion (degree-2 truncation)
C	$k = c_3/2 = \frac{1}{16\pi}$, $S/A = \frac{1}{4}$	**survives**	absolute $1/G$ (UV-sensitive) is the one anchor
D	ratios+products ⇒ one scale v_{geo}	**survives (narrowed)**	CP phases ($\delta_{\mathrm{CKM/PMNS}}$) not covered by v_{geo}
E	one scale v_{geo} carries the theory	**survives (narrowed)**	EW/reheating/leptogenesis scales + seam=Planck identification

Method

The red team treats each reduction as a hostile witness. A confirmatory script that always passes is worthless here: the layer is built so that it *may* downgrade a claim. Three outcomes are allowed — **survives** (stands as worded), **survives (narrowed)** (stands only after a silent assumption is made explicit), and **reduced, not closed** (the conservative wording is the correct one). A fourth, **broken**, is reserved for an actual failure; none occurred.

Target A — the $(E8)_1$ boundary-net identification

Minimal claim. The ambient metric/projective measure is reduced to identifying the seam–Calderón boundary measure with the $(E8)_1$ lattice net, carrier subnet $(D5)_1 \times (A3)_1$. [`verification/redteam/rt_A_e8net.py`]

Logical chain (established). The Sugawara level-1 central charges are $c(E8)_1 = \frac{248}{31} = 8$, $c(D5)_1 = 5 = g_{\mathrm{car}}$, $c(A3)_1 = 3 = N_{\mathrm{fam}}$, so the conformal-embedding criterion $c_{\mathrm{coset}} = c(E8) - c(D5) - c(A3) = 0$ holds [I]. This is *compatibility*.

Where it breaks: central charge underdetermines the net

Counterexample: $(D8)_1 = SO(16)_1$ also has $c = \frac{120}{15} = 8$, but is *not* holomorphic (four primaries $1, v, s, c$) — a distinct $c = 8$ net. Hence equal central charge is *necessary, not sufficient*: $(E8)_1$ is the unique *holomorphic* $c = 8$ net (even unimodular rank-8 lattice $= E8$), so **holomorphy (single vacuum sector, μ-index 1) is the load-bearing extra assumption**. Dropping chirality is worse: the $c = 8$ Narain family is positive-dimensional. The constructive map seam–Calderón kernel $\to (E8)_1$ and the uniqueness of bulk reconstruction are not established; the gap $\Delta_{\mathrm{eff}} > 0$ *supports* tightness (clustering) but does not prove it.

Verdict. Keep "reduced to a rigorous boundary-net identification problem"; do *not* write "metric sector closed". The red team confirms the conservative wording. Status [P]. Residual: holomorphy proof + constructive boundary map + bulk-reconstruction uniqueness.

Target B — carrier rank / the Pascal condition

Minimal formal claim. $2^g = g^2 + g + 2$ has the unique solution $g = 5$ (Lean-verified [F]). This arithmetic is *not* attacked. The attack is upstream: why must the carrier obey the Pascal half-spinor exhaustion $2^{g-1} = \binom{g}{0} + \binom{g}{1} + \binom{g}{2}$? [`verification/redteam/rt_B_pascal.py`]

Where it narrows: the rule, not the arithmetic, selects 5

Perturbing the constant breaks it: $2^g = g^2 + g + c$ gives $\{5\}$ only at $c = 2$; neighbours give other or empty solution sets. The truncation *degree* is the real choice: with $2^{g-1} = \sum_{k \le K} \binom{g}{k}$ one finds $K{=}0 \to g{=}1$, $K{=}1 \to 3$, $K{=}2 \to 5$, $K{=}3 \to 7$, i.e. $g = 2K + 1$. So choosing $K = 2$ (to reach $g_{\mathrm{car}} = 5$) is a physical postulate. It is corroborated — the $D5$ half-spinor $\dim S^+ = 2^{g-1} = 16 \Leftrightarrow g = 5$ and the family closure $N_{\mathrm{fam}} = (2^{g-1}{-}1)/g$, $g + N_{\mathrm{fam}} = 8$ both pick 5 — but each is itself a postulate, not a theorem.

Verdict. Theorem A's arithmetic core stands; the Pascal *selection* must be typed [A]/[P], not [I]. Residual: a boundary/seam derivation of half-spinor exhaustion.

Target C — $k = c_3/2$ and the seam-area coefficient

Minimal claim. In reduced units $k = c_3/2 = \frac{1}{16\pi}$ and Fursaev–Solodukhin gives $S/A = \frac{1}{4}$. [`verification/redteam/rt_C_kc3.py`]

The dimensional firewall holds

The only legal chain is

$$k_{\mathrm{red}} = \frac{c_3}{2} = \frac{1}{16\pi} \ \text{(dimensionless)}, \qquad k_{\mathrm{phys}} = \frac{k_{\mathrm{red}}}{G}, \qquad S = 4\pi\, k_{\mathrm{phys}} A_{\mathrm{phys}} = \frac{A_{\mathrm{phys}}}{4G}.$$

The forbidden form $k_{\mathrm{phys}} = c_3/2$ is dimensionally false (legal only at $G = 1$). A scan of `v67/v68/v73` finds every $c_3/2$ used as the *reduced* coefficient and **zero** naked dimensionful identities; the firewall's teeth are verified against a synthetic violation. Internally consistent with $1/G$ being UV-sensitive (Sakharov/Connes, v68).

Verdict. Safe as worded (reduced coefficient). Status [I] (structure) + [A] for the residual. Residual: the absolute $4D$ Newton scale $1/G$ ($\propto \Lambda^2 f_2$) stays the one dimensionful anchor; nothing *dimensionless* is open.

Target D — $U_{\text{point}} \to v_{\text{geo}}$

Minimal claim. Ratios plus sector products determine the dimensionless amplitudes, leaving one common scale v_{geo}. [`verification/redteam/rt_D_upoint.py`]

Where it narrows: the bijection needs explicit hypotheses

For positive reals, (ratios, product) $\Leftrightarrow$ (individuals) is a bijection, and the TFPT sectors satisfy it (positive rational c, odd size $3 = N_{\text{fam}}$). But it *fails* off-assumption: even sector size leaves a sign ambiguity ($\{2,3\}$ vs $\{-2,-3\}$ share ratio 3/2 and product 6); complex amplitudes leave an n-th-root branch ambiguity ($\{1,1,1\}, \{\omega,\omega,\omega\}, \{\omega^2,\omega^2,\omega^2\}$ share all ratios and product 1); a zero amplitude is degenerate. **Genuine residual:** the bijection fixes amplitude *magnitudes* only — CP phases $\delta_{\text{CKM}}, \delta_{\text{PMNS}}$ are complex and are *not* folded into v_{geo}.

Verdict. Holds for TFPT once four hypotheses (real, positive, fixed branch, no zeros / odd sector size) are made explicit; they are silent in v75 and must be named. Status [I] (reduction) + [A] (v_{geo}). Residual: the CP-phase sector.

Target E — v_{geo} as the unique dimensional floor

Minimal claim. No pure-number theory can derive an absolute dimensionful scale; all scales are ratios to one unit v_{geo}. [`verification/redteam/rt_E_vgeo.py`]

Single-scale audit: exact for the certified tiers, conditional beyond

The closed compiler + protected IR readouts all reduce to (pure number)$\times v_{\text{geo}}$: $M_{\text{scal}} = c_3^{7/2}\bar{M}_{\text{Pl}}$, $f_a = (c_3/|\mu_4|)^{7/2}\bar{M}_{\text{Pl}}$, masses $= (\pi/\sqrt{2})\, c_f\, \varphi_0^k\, v_{\text{geo}}$, and $1/G$ shares the same anchor (v68/v75). The audit *surfaces* candidates that are not pure reductions: the seam cutoff Λ (only an *identification* seam=Planck collapses it to v_{geo}), the electroweak matching scale, the reheating temperature, and the leptogenesis scale — all frontier [P]/[A] inputs that must not be silently absorbed into v_{geo}.

Verdict. v_{geo} is the unique dimensional floor of the certified tiers (true by dimensional analysis). Full single-scale-ness is conditional on the flagged identifications staying honestly typed. Status [I] (ratios) + [A] (one scale). Residual: explicit reduction (or honest frontier typing) of the EW / reheating / leptogenesis scales + the seam=Planck cutoff.

Outcome

What the red team changed

No target broke. The hardest gate (A) is confirmed *open* with the conservative wording; three targets (B, D, E) *survive once a silent assumption is named*; one (C) survives as worded. The net effect is not new physics but sharper typing: the package now states *where* each reduction would fail and *which* assumptions are truly necessary, exactly at the weakest logical transitions. Reproduce with `cd verification/redteam && python run_redteam.py`.

Follow-up — the verdicts were then used as a worklist [L]/[I] ([verification/v83_e8net_holomorphic_uniqueness.py])

Two residuals were attacked directly after the red-team pass; the result is recorded in ledger rows `GATE.METRIC.04` and `CAR.PASCAL.01`.

Target A, residual 1 — closed [L]. At level 1 the number of primaries equals det(Cartan) = |fundamental group|: A_3=4, D_5=4, D_8=4, $\mathbf{E_8}$=**1**. So holomorphy (single vacuum sector, μ-index 1) is *necessary* — it excludes the same-c rival $(D_8)_1 = SO(16)_1$ ($c = 120/15 = 8$ but four primaries $1, v, s, c$) — *and sufficient*: a holomorphic $c = 8$ chiral CFT is the lattice theory of an even unimodular rank-8 lattice, of which there is exactly one, E_8, by the Minkowski–Siegel mass formula (mass $= 1/|W(E_8)| = 1/696729600$, and $|\mathrm{Aut}(E_8)|$ saturates it). The $(D_5)_1 \times (A_3)_1$ embedding $c = 5+3 = 8$ stays *compatibility* (16 primaries vs 1). **Consequence:** the constructive map need only show the seam–Calderón boundary net is holomorphic with $c = 8$ (then E_8 is automatic), so Target A drops from *three* residuals to *two*: (i) boundary-net holomorphy $+\, c = 8$ ($\Delta_{\text{eff}} > 0$ supports, not proves), and (ii) bulk-reconstruction uniqueness — both still [P]/[A].

Target B — reduced [I]/[L]. The "$K = 2$ Pascal truncation" is not free: $K = (g-1)/2$ is the Pascal-row midpoint $\sum_{k\le(g-1)/2}\binom{g}{k} = 2^{g-1}$ (odd g), i.e. the half-spinor split; the carrier is the even Clifford half-spinor $\Lambda^{\text{even}}(\mathbb{C}^5) = 1 + 10 + 5 = 16 = \dim S^+$. So B's residual reduces to the single standard input "carrier = half-spinor of Spin(10)" (the Lean arithmetic core stays [F]).

Still open, typed honestly: A (i)+(ii); Target D's CP-phase residual (at $\theta_{13} = 0$ the Dirac phase is undefined, so this needs a texture correction); Target E's reheating/leptogenesis scales and the seam=Planck identification. Target C is a dimensional *floor*, not a gap. **No target was promoted.**

Second follow-up round [L]/[N] ([verification/v87_bulk_uniqueness_reduction.py], [verification/v88_cp_phase_audit.py], [verification/v86_nstar_reheating.py])

Target A, residual (ii) — conditionally closed [L]: Target A = ONE residual. Bulk-reconstruction uniqueness is *not* independent of residual (i). By the algebraic-QFT classification of full 2D CFTs over a chiral net (Longo–Rehren; Kawahigashi–Longo–Müger; Bischoff–Kawahigashi–Longo–Rehren), the possible bulks are the haploid commutative Q-systems $\simeq$ physical modular invariants of Rep(A). For a *holomorphic* net Rep(A) = Vect, so the bulk pairing is *unique*. The contrast is machine-checked: the same-c rival $SO(16)_1$ (discriminant category $\mathbb{Z}_2\times\mathbb{Z}_2$, μ-index 4) admits *six* modular invariants — diagonal, $s\leftrightarrow c$ swap, both E_8-extension invariants $|\chi_0+\chi_{s/c}|^2$ (the lattice glue $E_8 = D_8 \cup (D_8+s)$, h_s=1 bosonic), and two twisted pairings — so without holomorphy the $c = 8$ bulk is *multiply* ambiguous; with it, trivially unique. Hence the entire Target A reduces to the single theorem "seam–Calderón boundary net is holomorphic with $c = 8$" ([P]/[A]; ledger `GATE.METRIC.05`).

Target D — the CP-phase residual quantified [N] (not closed). With the frozen leading reading $\delta = \pi/3 + 3\lambda_C^2 = 68.65°$: pull vs $\gamma_{\text{PDG}} = 65.7° \pm 3.0°$ is $+0.98\sigma$ (survives). Audit, *not* promoted: the data central value sits 0.07° from the alternative coefficient reading $\pi/3 + 2\lambda_C^2$ ($|\mathbb{Z}_2|$ instead of N_{fam}) — a textbook look-elsewhere trap; the registry keeps coefficient 3 (`REG.FREEZE.01`). Decision threshold: the readings differ by $\lambda_C^2 = 2.88°$, i.e. 3σ-distinguishable once $\sigma_\gamma \le 0.96°$. The J-inversion is flagged magnitude-contaminated (source-scale s_{23} vs M_Z). Ledger `FLAV.CP.01`.

Target E — one flagged scale computed [P]. The reheating temperature is no longer a silent input: $M_{\text{scal}} = c_3^{7/2}\bar{M}_{\text{Pl}}$ [I] + standard physics give $T_{\text{reh}} = 9.6 \times 10^9$ GeV and $N_\star(0.05/\text{Mpc}) = 51.4$ [P] (ledger `COSMO.NSTAR.01`); honestly recorded: this slow Higgschannel point is A_s-*disfavoured* (-11.4σ; the measured A_s requires near-instantaneous reheating), so the point is conditional on the decay channel and the frozen band $[50, 60]$ stays the surface of record; the leptogenesis scale and the seam=Planck identification remain typed inputs. **Again: no target was promoted; one residual was *merged* (A), one was *quantified* (D), one input was *computed* (E).**